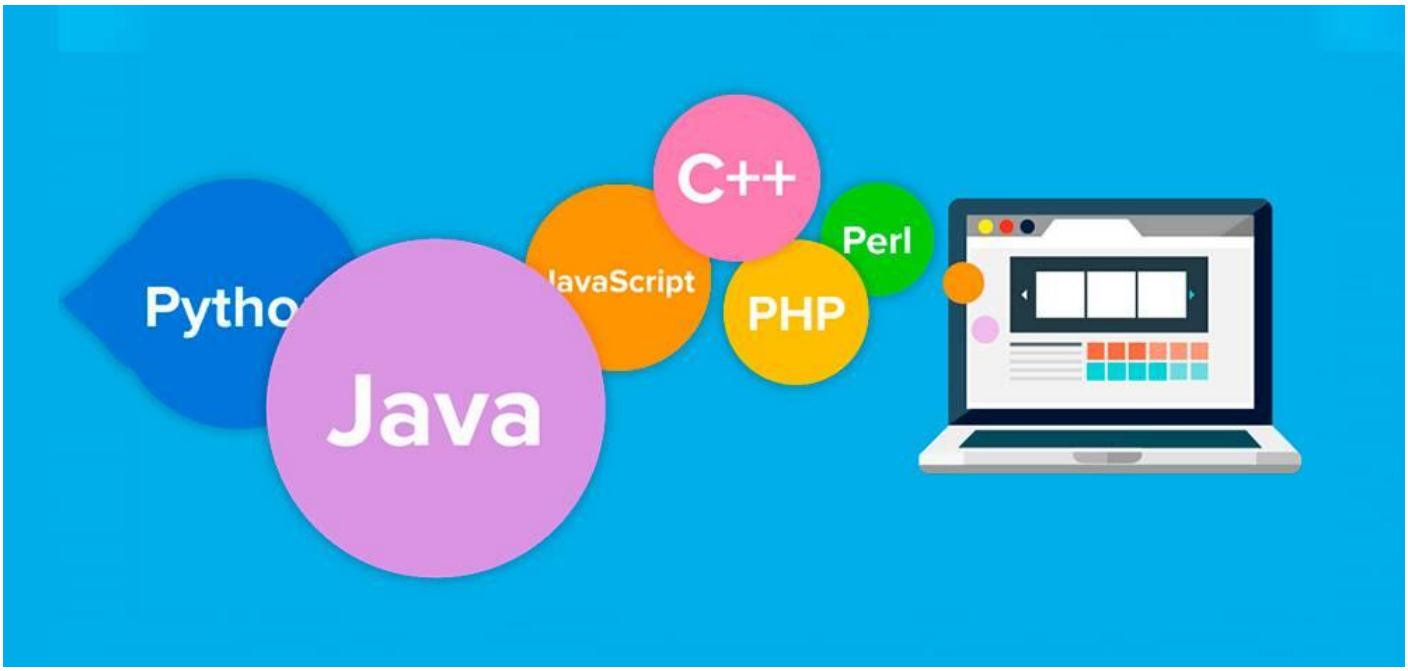
University of the Armed Forces

DEPARTMENT OF COMPUTER SCIENCE

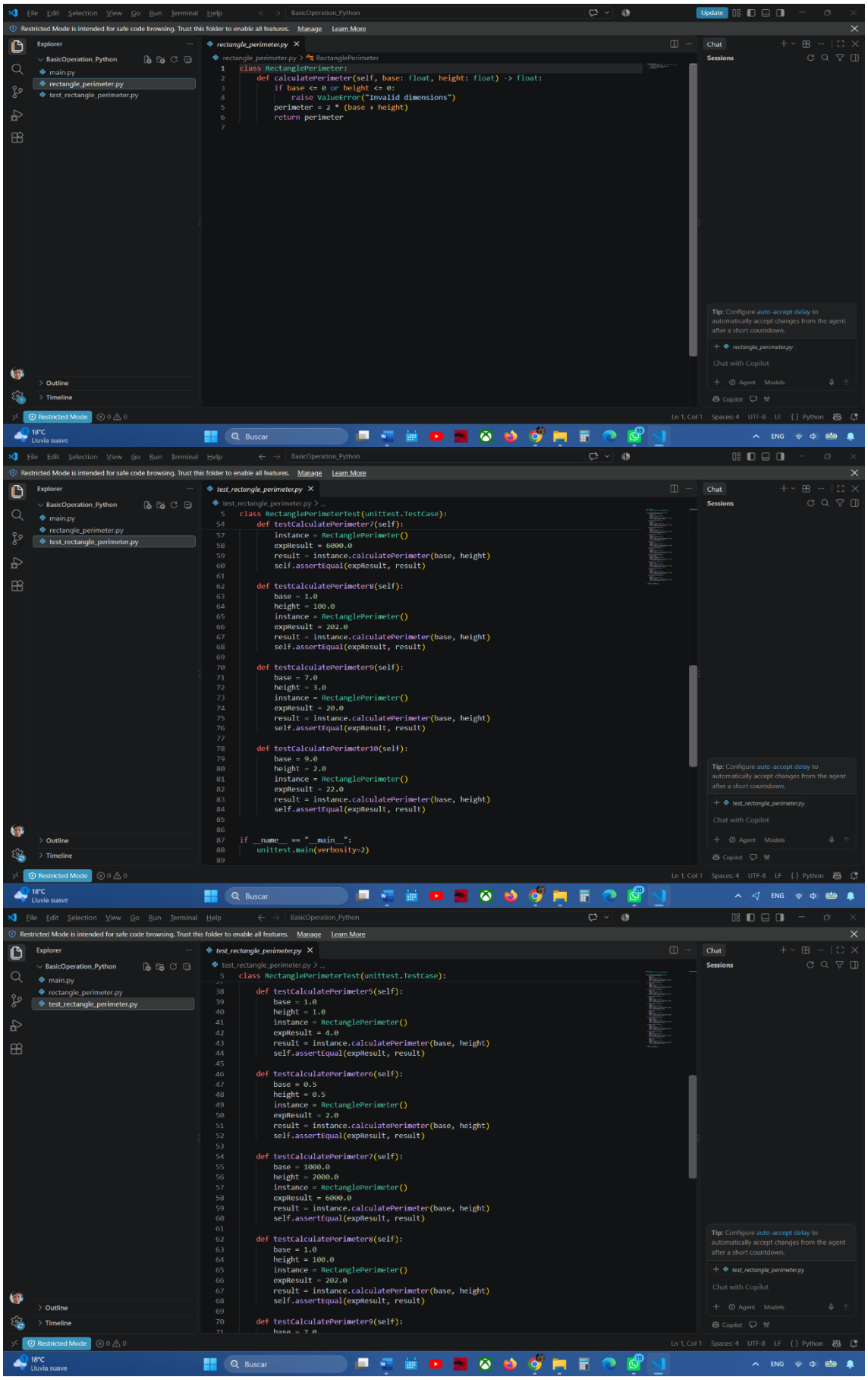
Object-Oriented Programming

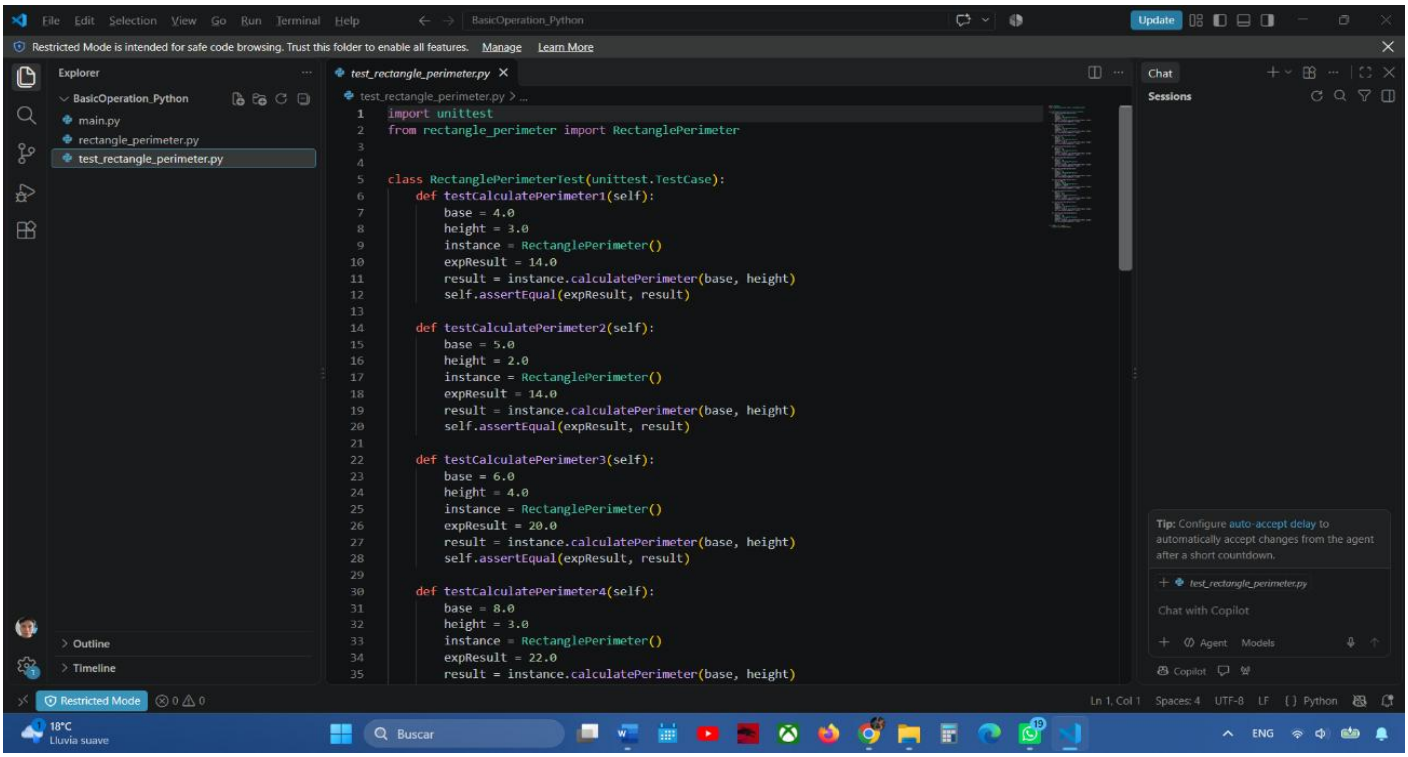
Homework 14: Files

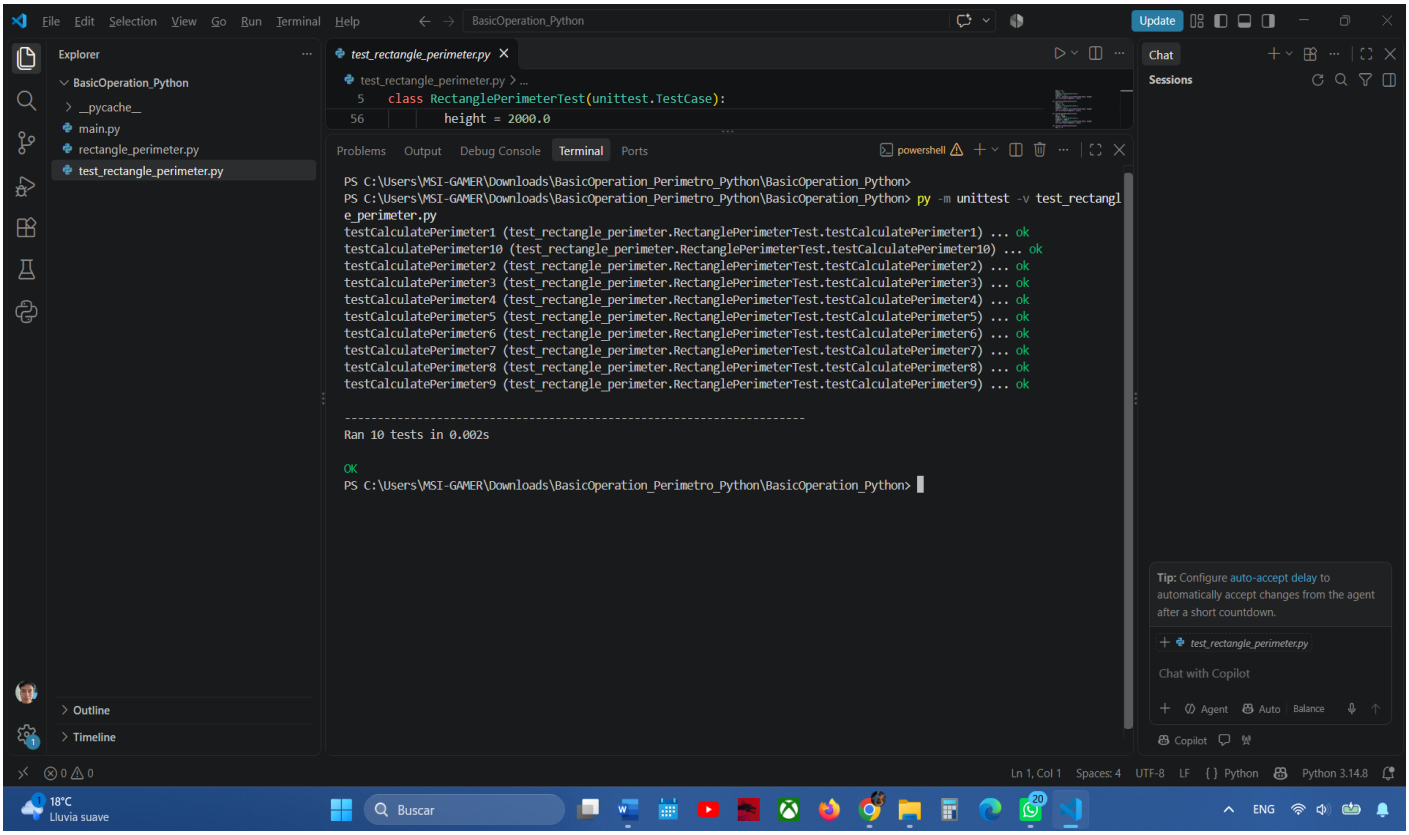
Edison Lascano PhD
NRC: 36862



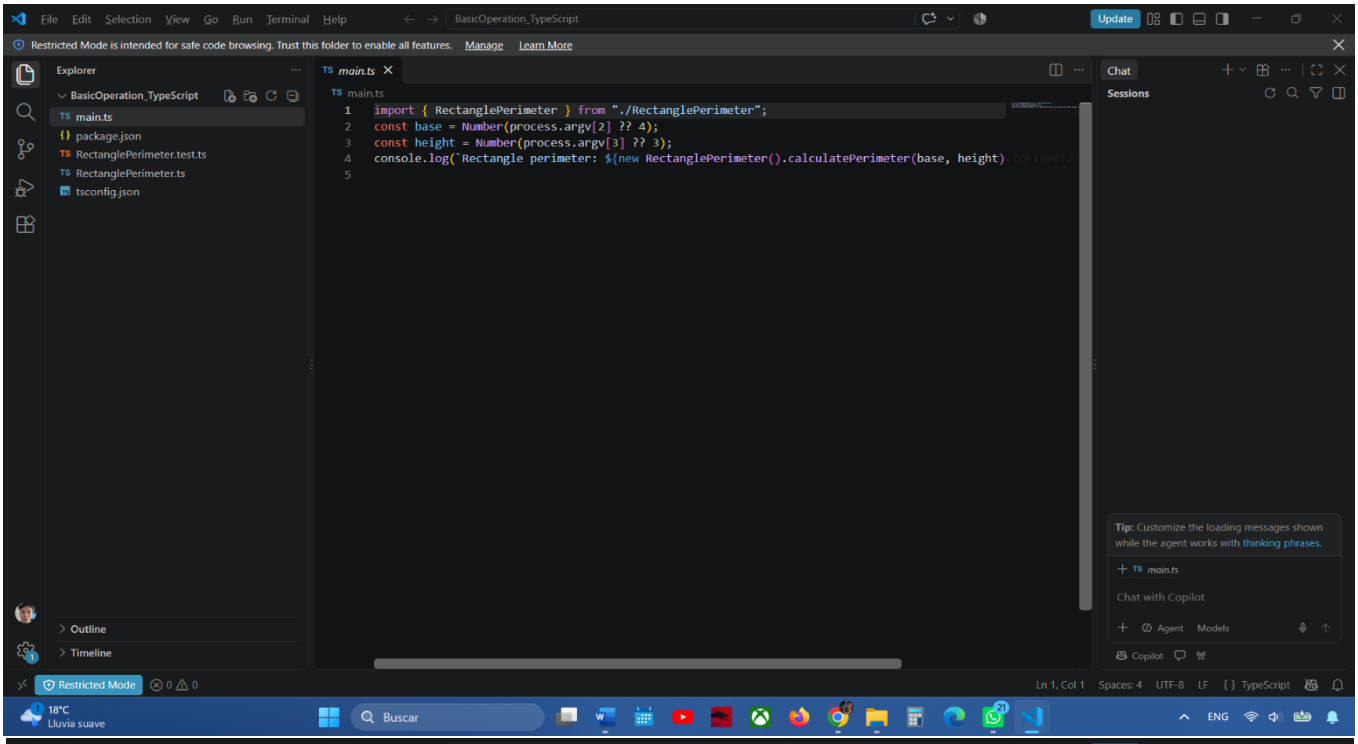
Python

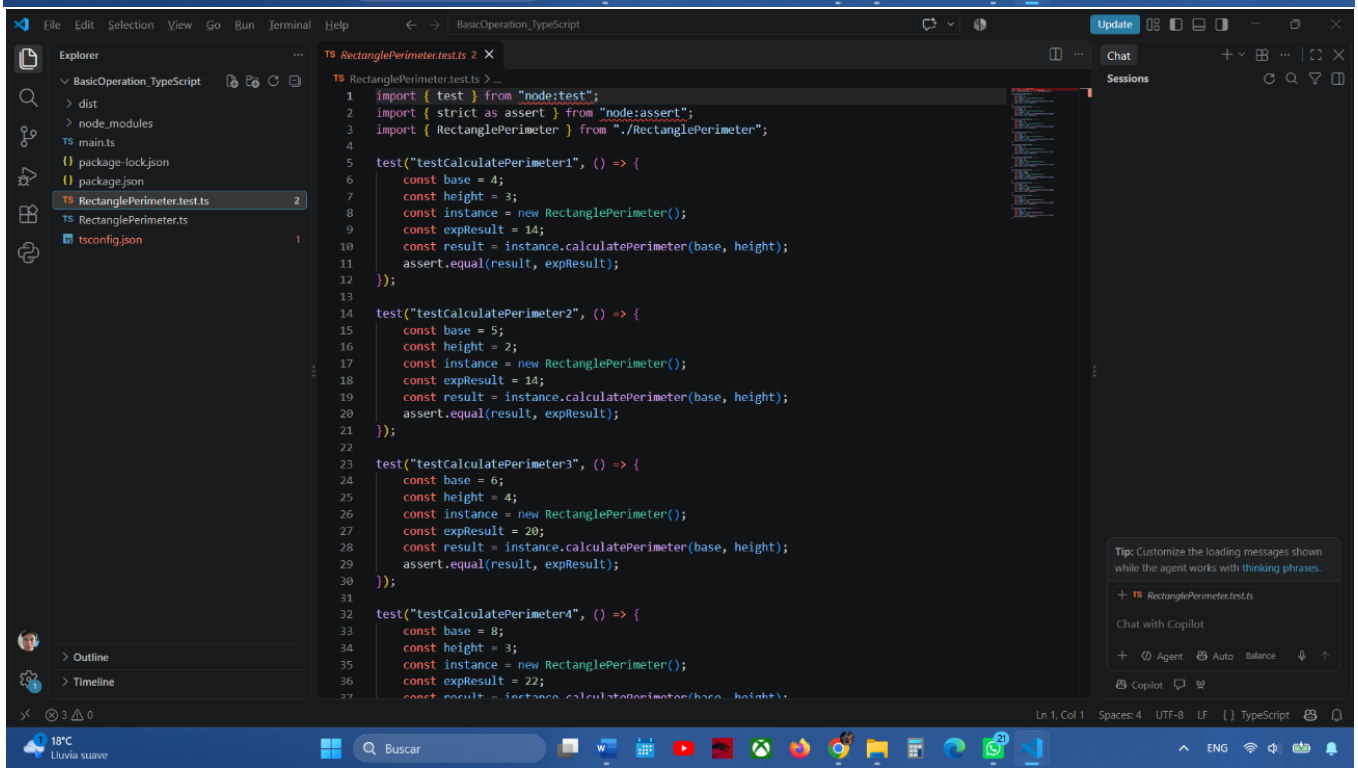

rectangle_perimeter.py
class RectanglePerimeter:
def calculatePerimeter(self, base: float, height: float) -> float:
if base <= 0 or height <= 0:
raise ValueError("invalid dimensions")
perimeter = 2 * (base + height)
return perimeter
test_rectangle_perimeter.py
class RectanglePerimeterTest(unittest.TestCase):
def testCalculatePerimeter7(self):
instance = RectanglePerimeter()
expResult = 6000.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter8(self):
base = 1.0
height = 100.0
instance = RectanglePerimeter()
expResult = 202.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter9(self):
base = 7.0
height = 3.0
instance = RectanglePerimeter()
expResult = 20.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter10(self):
base = 9.0
height = 2.0
instance = RectanglePerimeter()
expResult = 22.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
if __name__ == "__main__":
unittest.main(verbosity=2)
def testCalculatePerimeter5(self):
base = 1.0
height = 1.0
instance = RectanglePerimeter()
expResult = 4.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter6(self):
base = 0.5
height = 0.5
instance = RectanglePerimeter()
expResult = 2.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter7(self):
base = 1000.0
height = 2000.0
instance = RectanglePerimeter()
expResult = 6000.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter8(self):
base = 1.0
height = 100.0
instance = RectanglePerimeter()
expResult = 202.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter9(self):

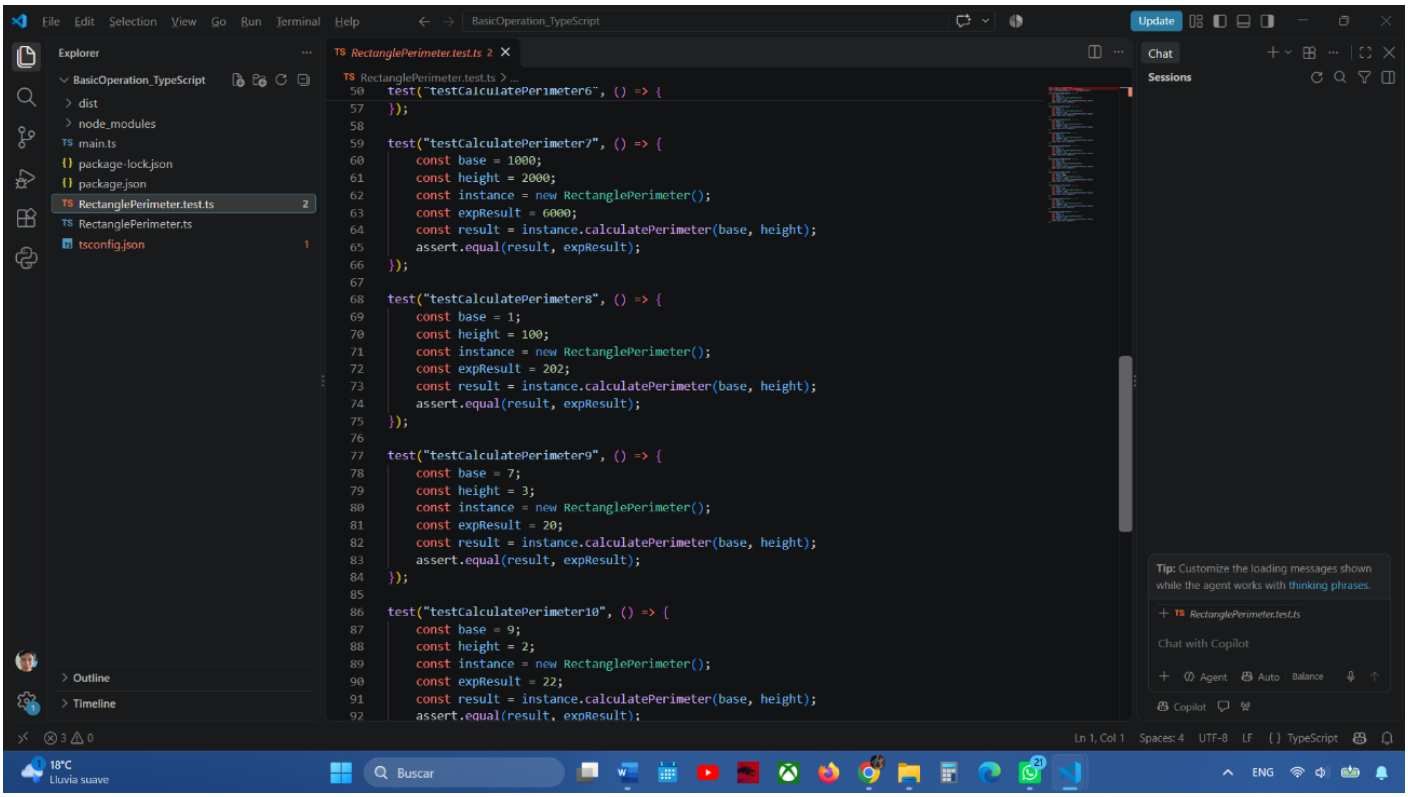

File Edit Selection View Go Run Terminal Help
BasicOperation_Python
Update
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
Explorer
BasicOperation_Python
main.py
rectangle_perimeter.py
test_rectangle_perimeter.py
test_rectangle_perimeter.py
import unittest
from rectangle_perimeter import RectanglePerimeter
class RectanglePerimeterTest(unittest.TestCase):
def testCalculatePerimeter1(self):
base = 4.0
height = 3.0
instance = RectanglePerimeter()
expResult = 14.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter2(self):
base = 5.0
height = 2.0
instance = RectanglePerimeter()
expResult = 14.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter3(self):
base = 6.0
height = 4.0
instance = RectanglePerimeter()
expResult = 20.0
result = instance.calculatePerimeter(base, height)
self.assertEqual(expResult, result)
def testCalculatePerimeter4(self):
base = 8.0
height = 3.0
instance = RectanglePerimeter()
expResult = 22.0
result = instance.calculatePerimeter(base, height)
Chat
Sessions
Tip: Configure auto-accept delay to automatically accept changes from the agent after a short countdown.
test_rectangle_perimeter.py
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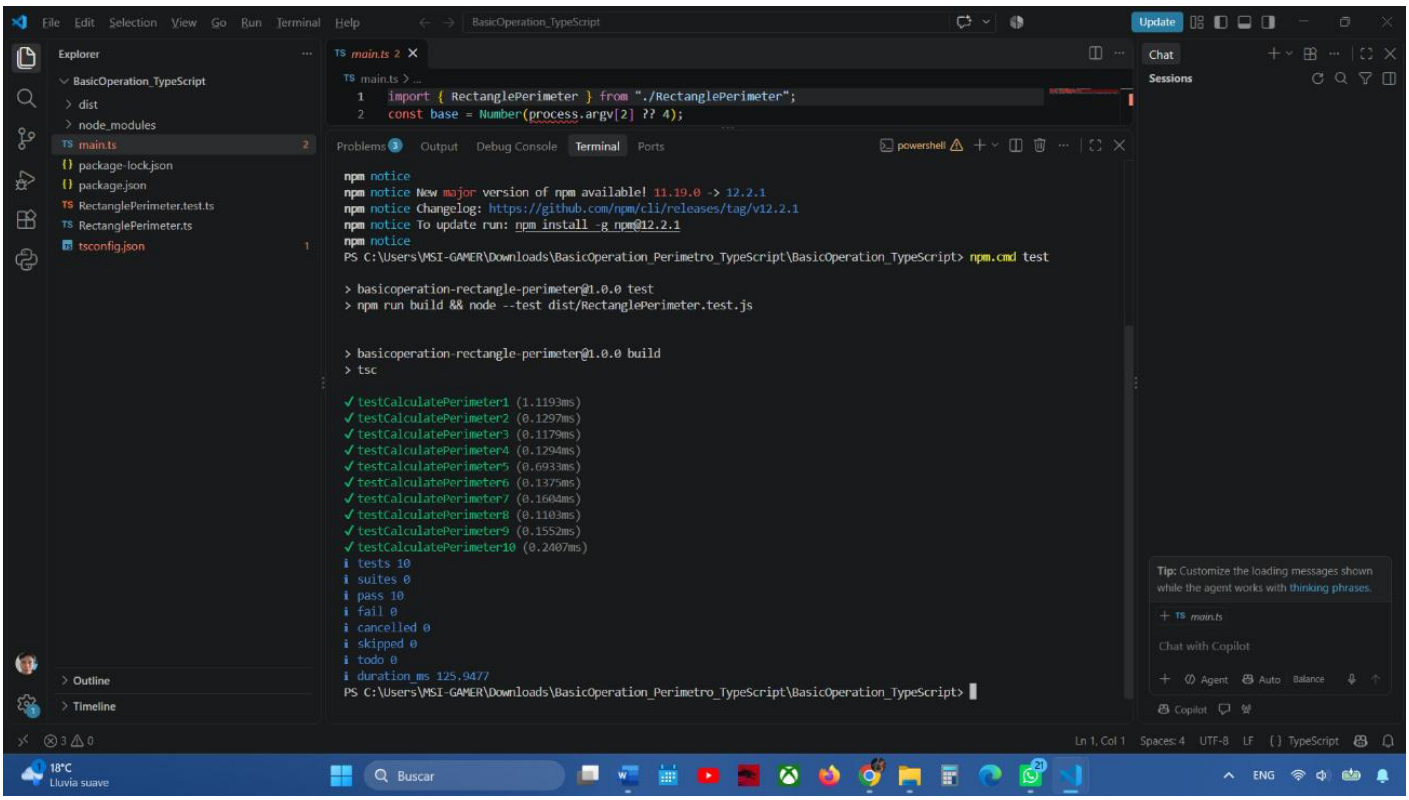

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BasicOperation_Python
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BasicOperation_Python
__pycache__
main.py
rectangle_perimeter.py
test_rectangle_perimeter.py
test_rectangle_perimeter.py
class RectanglePerimeterTest(unittest.TestCase):
height = 2000.0
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\MSI-GAMER\Downloads\BasicOperation_Perimetro_Python\BasicOperation_Python>
PS C:\Users\MSI-GAMER\Downloads\BasicOperation_Perimetro_Python\BasicOperation_Python> py -m unittest -v test_rectangle_perimeter.py
testCalculatePerimeter1 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter1) ... ok
testCalculatePerimeter10 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter10) ... ok
testCalculatePerimeter2 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter2) ... ok
testCalculatePerimeter3 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter3) ... ok
testCalculatePerimeter4 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter4) ... ok
testCalculatePerimeter5 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter5) ... ok
testCalculatePerimeter6 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter6) ... ok
testCalculatePerimeter7 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter7) ... ok
testCalculatePerimeter8 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter8) ... ok
testCalculatePerimeter9 (test_rectangle_perimeter.RectanglePerimeterTest.testCalculatePerimeter9) ... ok
Ran 10 tests in 0.002s
OK
PS C:\Users\MSI-GAMER\Downloads\BasicOperation_Perimetro_Python\BasicOperation_Python>
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Tip: Configure auto-accept delay to automatically accept changes from the agent after a short countdown.
test_rectangle_perimeter.py
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TypeScript


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BasicOperation_TypeScript
Update
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BasicOperation_TypeScript
main.ts
package.json
RectanglePerimeter.test.ts
RectanglePerimeter.ts
tsconfig.json
main.ts
import { RectanglePerimeter } from "./RectanglePerimeter";
const base = Number(process.argv[2] ?? 4);
const height = Number(process.argv[3] ?? 3);
console.log(`Rectangle perimeter: ${new RectanglePerimeter().calculatePerimeter(base, height)
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Tip: Customize the loading messages shown while the agent works with thinking phrases.
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package-lock.json
package.json
RectanglePerimeter.test.ts
RectanglePerimeter.ts
tsconfig.json
RectanglePerimeter.test.ts
import { test } from "node:test";
import { strict as assert } from "node:assert";
import { RectanglePerimeter } from "./RectanglePerimeter";
test("testCalculatePerimeter1", () => {
const base = 4;
const height = 3;
const instance = new RectanglePerimeter();
const expResult = 14;
const result = instance.calculatePerimeter(base, height);
assert.equal(result, expResult);
});
test("testCalculatePerimeter2", () => {
const base = 5;
const height = 2;
const instance = new RectanglePerimeter();
const expResult = 14;
const result = instance.calculatePerimeter(base, height);
assert.equal(result, expResult);
});
test("testCalculatePerimeter3", () => {
const base = 6;
const height = 4;
const instance = new RectanglePerimeter();
const expResult = 20;
const result = instance.calculatePerimeter(base, height);
assert.equal(result, expResult);
});
test("testCalculatePerimeter4", () => {
const base = 8;
const height = 3;
const instance = new RectanglePerimeter();
const expResult = 22;
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package-lock.json
package.json
RectanglePerimeter.test.ts 2
RectanglePerimeter.ts
tsconfig.json 1
RectanglePerimeter.test.ts 2
RectanglePerimeter.test.ts > ...
50 test("testCalculatePerimeter6", () => {
57 });
58
59 test("testCalculatePerimeter7", () => {
60 const base = 1000;
61 const height = 2000;
62 const instance = new RectanglePerimeter();
63 const expResult = 6000;
64 const result = instance.calculatePerimeter(base, height);
65 assert.equal(result, expResult);
66 });
67
68 test("testCalculatePerimeter8", () => {
69 const base = 1;
70 const height = 100;
71 const instance = new RectanglePerimeter();
72 const expResult = 202;
73 const result = instance.calculatePerimeter(base, height);
74 assert.equal(result, expResult);
75 });
76
77 test("testCalculatePerimeter9", () => {
78 const base = 7;
79 const height = 3;
80 const instance = new RectanglePerimeter();
81 const expResult = 20;
82 const result = instance.calculatePerimeter(base, height);
83 assert.equal(result, expResult);
84 });
85
86 test("testCalculatePerimeter10", () => {
87 const base = 9;
88 const height = 2;
89 const instance = new RectanglePerimeter();
90 const expResult = 22;
91 const result = instance.calculatePerimeter(base, height);
92 assert.equal(result, expResult);
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Tip: Customize the loading messages shown while the agent works with thinking phrases.
RectanglePerimeter.test.ts
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node_modules
main.ts 2
package-lock.json
package.json
RectanglePerimeter.test.ts
RectanglePerimeter.ts
tsconfig.json 1
main.ts 2
main.ts > ...
1 import { RectanglePerimeter } from "./RectanglePerimeter";
2 const base = Number(process.argv[2] ?? 4);
Problems 3 Output Debug Console Terminal Ports
powershell
npm notice
npm notice New major version of npm available! 11.19.0 -> 12.2.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.2.1
npm notice To update run: npm install -g npm@12.2.1
npm notice
PS C:\Users\MSI-GAMER\Downloads\BasicOperation_Perimetro_TypeScript\BasicOperation_TypeScript> npm.cmd test
> basicoperation-rectangle-perimeter@1.0.0 test
> npm run build && node --test dist/RectanglePerimeter.test.js
> basicoperation-rectangle-perimeter@1.0.0 build
> tsc
✓ testCalculatePerimeter1 (1.1193ms)
✓ testCalculatePerimeter2 (0.1297ms)
✓ testCalculatePerimeter3 (0.1179ms)
✓ testCalculatePerimeter4 (0.1294ms)
✓ testCalculatePerimeter5 (0.6933ms)
✓ testCalculatePerimeter6 (0.1375ms)
✓ testCalculatePerimeter7 (0.1604ms)
✓ testCalculatePerimeter8 (0.1103ms)
✓ testCalculatePerimeter9 (0.1552ms)
✓ testCalculatePerimeter10 (0.2407ms)
ℹ tests 10
ℹ suites 0
ℹ pass 10
ℹ fail 0
ℹ cancelled 0
ℹ skipped 0
ℹ todo 0
ℹ duration_ms 125.9477
PS C:\Users\MSI-GAMER\Downloads\BasicOperation_Perimetro_TypeScript\BasicOperation_TypeScript>
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